

Abstract No.

Integrated Spatial Profiling Identifies a Recruitment-Deficient Lymphocyte-Cold State in Meningioma

Klejdi Nurellari^{1,2}, Aso Omer Mohammed^{1,2}, Mohammad Al Shhab^{1,2}, Michel Goth^{1,2}, Felix Sahm³, Tareq Juratli⁴, Jürgen Beck^{1,2}, Jürgen Grauvogel^{1,2}, Kevin Joseph^{1,2}, and Vidhya M Ravi^{1,2}

¹Klinik für Neurochirurgie, Universitätsklinikum Freiburg

²Medizinische Fakultät

³Institut of Neuropathologie, Universitätsklinikum Heidelberg

⁴Klinik für Neurochirurgie, Universitätsklinikum Dresden

Integrated Spatial Profiling Identifies a Recruitment-Deficient Lymphocyte-Cold State in Meningioma

Background: Immune checkpoint blockade has shown limited efficacy in meningioma, yet the mechanisms underlying immune resistance remain unclear. While hypoxic, macrophage-rich niches have been implicated in immune evasion, it remains unresolved whether lymphocyte-cold tumors result from local T-cell exclusion, T-cell dysfunction, or defective immune recruitment. We integrated transcriptomic, spatial, and protein-level profiling to define the meningioma immune microenvironment and established a patient-derived slice-culture platform for functional validation.

Methods: Bulk RNA-seq from 273 meningiomas was integrated with multiplex immunofluorescence (42 WHO grade I–II tumors), spatial transcriptomics (29 Visium sections), and single-nucleus RNA-seq (26 tumors; 44,266 cells). Spatial autocorrelation and neighborhood enrichment analyses quantified relationships between T cells, hypoxia, and macrophage-rich niches. Patient-derived organotypic slice cultures co-cultured with matched autologous PBMCs or isolated CD8 T cells were established for mechanistic studies using live-cell imaging, multiplex immunofluorescence, flow cytometry, and CXCL9/CXCL10 ELISA.

Results: Bulk transcriptomics identified reproducible immune microenvironment states independent of tumor grade and associated with coordinated hypoxic, angiogenic, and macrophage-related programs. Multiplex immunofluorescence confirmed organized CA9⁺/SPP1⁺/CD163⁺ macrophage-rich niches. Despite prominent macrophage clustering, CD3⁺ T cells remained diffusely distributed and were not depleted from hypoxic macrophage-rich regions, arguing against local macrophage-mediated exclusion as the mechanism underlying the lymphocyte-cold phenotype. Across bulk, spatial, and single-nucleus datasets, lymphocyte-cold tumors demonstrated impaired T-cell recruitment programs, characterized by reduced CXCL9–CXCR3 signaling despite preserved myeloid infiltration. Patient-derived slice cultures with matched autologous immune cells enabled assessment of T-cell recruitment into intact tumor tissue and provide a platform to perturb the CXCL9/CXCR3 axis.

Conclusions: Integrated spatial profiling identifies lymphocyte-cold meningiomas as a recruitment-deficient immune state rather than one driven by local T-cell exclusion. These findings provide a mechanistic framework for the limited efficacy of checkpoint blockade and nominate restoration of CXCL9–CXCR3-mediated T-cell recruitment as a therapeutic strategy.



Supported by SciSerTec

Web: <http://www.scisertec.de>

